

Assignment 2

Team Name: Panem

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Instructions

- Please **work in your teams**.
- Please go to Assignment 2 on the Assignments page in Canvas and use the upload link to submit your answers in a PDF file named **TEAMNAME_A2.pdf**.
- To receive full credit, please show your calculations for quantitative questions and explain your reasoning for qualitative questions.

Part 1: Trading

You can do most of this section in our Monday night game session.

These questions are related to the Trading Game. Use the Trading Game link posted on Canvas to access the game.

For the set of questions below, assume that you are operating in an OPEN ECONOMY, where you engage in trade with others.

Scenario information:

- The objective in the Trading Game are to maximize cash, practice trading with other teams, and optimize wholesale prices and supply chains. We will change our objectives later in the more advanced games.
- Each company has 2 products: one hormone product that all teams have and one proprietary specialty product.
- All products cost \$100 per unit to manufacture.
- All products are produced in batches of 100 units.
- It takes 2.5 days to produce a batch.
- It takes 24 hours to ship the batch from your factory to your distribution center (DC). The shipping within your region is free.
- Between regions, it takes 3 days and costs \$40/unit to ship by mail, and it takes 21 days and costs \$10,000 per container. Mail is shipped in sets of 10 units. Containers hold 1,000 units.
- The maximum willingness to pay (WTP) for a product is the same across all regions.
- The market size for the hormone product is 300,000 in each region.
- The market size for the specialty product in its home region is 140,000, but market size varies across other regions.
- All sales are made from your DC.
- Retail customers arrive your DC at the rate of $0.0001 \times \text{the market size per day}$.
- Potential retail customers either buy or don't buy. If they don't buy, they are lost forever. If there is no inventory when they come to the store, they are lost forever.
- Established shipping agreements will automatically reorder the agreed upon order quantity based on the inventory level and reorder point at the destination DC. To stop reordering, set the reorder point to -1.
- At the end of the game, all inventory becomes obsolete.

1. What is your production capacity, i.e., how many units does your factory produce each day?

180 divided by 2.5 = 40 units/day

2. What is your excess production capacity?

40 less expected sales of Hormone (12) minus expected sale of Speciality (7) = 21 units/day

3. How does the variation in market sizes across regions affect demand for specialty products and your negotiations with other teams? A qualitative answer is fine here.

Regions with smaller specialty markets have less bargaining power as buyers (lower volume = less attractive). Regions with large specialty markets are more valuable trade partners. Our negotiation leverage depends on how badly the counterparties need our specialty product vs how badly we need theirs.

4. What are some important considerations in determining the order quantity in shipping agreements with other teams?

For initial commencement, we will order based on the market demand of standard and specialized products in the home region and other markets. against the selling price (RSP) that we agreed with the wholesaler.

Subsequently, we will adjust our order based on Inventory Status report; A, stock on hand (days of inventory left), B, inventory stock out status

5. Write down the wholesale price that you think is appropriate to charge for each of your products when selling to other teams. Please do this even if you have decided to not sell a product. Explain why you believe this is the correct price to charge. Show your calculations.

Standard Product

- Our MC = \$100 (production). Shipping cost is \$40. Total cost to deliver = \$140/unit
- Optimal supply chain retail price = $(500+140)/2 = \$320$
- If you charge \$320 wholesale, buyer marks up to \$410 (double marginalization -> kills demand)
- Price range \$140 - \$320, and we sell at \$230 to split the margin about half with other teams

Specialty Product

- Our MC = \$100 (production). Shipping cost is \$40. Total cost to deliver = \$140/unit
 - Optimal supply chain retail price = $(1000 + 140)/2 = \$570$
 - If you charge \$570 wholesale, buyer marks up to \$785 (double marginalization -> kills demand)
 - Price range \$140 - \$570, and we sell at \$355 to split the margin about half with other teams
6. Write down the wholesale price that you think is appropriate to pay for each product bought from other teams. Explain why you believe this is the correct price to pay. Show your calculations.
- If we aim to maximise profit for a one-round game, we want to pay as close to the seller's MC (W = \$140 delivered) as possible.
 - I.e; for Hormone and Speciality = $\$100 + 40 = \140
 - However as this is a repeated game across the week, we will split the supply chain profit.
 - I.e; for Hormone = same as Q5 = \$230. Speciality = same as Q5 = \$355.
7. For both of your products, compare the unit contribution margins from wholesale and retail sales of that product. Please do this even if you have decided to not manufacture a product.

Contribution margin comparison;

Standard Product

- Retail margin (home, hormone at \$300): $\$300 - \$100 = \$200/\text{unit}$
- Wholesale margin (selling at say \$230): $\$230 - \$100 - \text{shipping by mail } \$40 = \$90/\text{unit}$
- Retail is ~2x more profitable per unit, but we have excess capacity. Therefore, it is ideal to sell the excess capacity at the wholesale arrangement.

Speciality Product

- Retail margin (home, hormone at \$550): $\$550 - \$100 = \$450/\text{unit}$
- Wholesale margin (selling at say \$355): $\$355 - \$100 - \text{shipping by mail } \$40 = \$215/\text{unit}$
- Retail is ~2x more profitable per unit, but we have excess capacity. Therefore, it is ideal to sell the excess capacity at the wholesale arrangement.

8. What is your strategy for allocating your limited production capacity? Explain.

Capacity allocation: Prioritize home region retail demand first (higher contribution margin), use excess capacity for wholesale to sell to other markets. Between wholesalers, prioritize regions with higher WTP.

9. When selling to other regions, how should you adjust the reorder point at your DC for products that you manufacture? A qualitative answer is fine here.

Based on actively selling to all other regions, we will sum up all the expected demand. Assume we priced RSP at \$570 for speciality item, we will capture 43% of the footfall for all other regions. Then we set the reorder point to the daily expected demand multiply by the production time plus lead time of delivery (add safety stock buffer too).

10. What analysis can you perform and what actions can you take to avoid stockouts?

Avoiding stock-outs: Monitor days-of-supply on hand, set reorder points above (lead time × daily demand), track the inventory status report, and keep a safety stock buffer.

Part 2: Supply Chain Profit

Please use this hypothetical information for the following set of questions. These questions are not related to any game scenario.

The seller is considering setting up a shipping agreement with another region (the buyer).

- Marginal production cost at the seller: \$100 / unit
- Marginal shipping cost incurred by the seller: \$40 / unit
- Retail customer maximum WTP: \$1,000 / unit
- Optimal retail price to maximize total supply chain profits:

$$\frac{\text{Max WTP} + \text{MC}}{2} = \frac{1,000 + 140}{2} = \$570$$

- 10 potential retail customers (footfall) per day at the buyer's DC

11. If the seller sells the product to the buyer for a wholesale price of \$570, what retail price maximizes the buyer's profits?

The buyer's optimal retail price works out to $(1000 + 570) \div 2 = \$785$.

12. What is the buyer's expected daily demand, if they price the good at the retail price calculated in (11)?

Demand at \$785 = $10 \times (1000 - 785) / 1000 = 2.15$ units/day

13. What is the buyer's daily profit, if they price the good at the retail price calculated in (11)?

Buyer profit = $2.15 \times (785 - 570) = \$462.25/\text{day}$

14. What is the daily supply chain profit (for all units sold to retail customers) if the seller sells the goods to the buyer at \$570 and they price at the retail price you calculated in (11)?

Supply chain profit = $2.15 \times (785 - 140) = \$1,386.75/\text{day}$

15. Referencing the profit that you calculated in (14), how is that profit split between the seller and the buyer in dollar terms?

The profit splits as seller \$924.50, buyer \$462.25.

16. What is the buyer's expected daily demand, if they price the good at the retail price of \$570?

At a \$570 retail price, demand is $10 \times 0.43 = 4.3 \text{ units/day}$.

17. What is the daily supply chain profit if the buyer sells the goods at a retail price of \$570?

The supply chain profit at a \$570 retail price is calculated as $4.3 \times (570 - 140) = \$1,849$ per day.

18. Referencing the profit calculated in (17), how is that profit split between the seller and the buyer if the buyer sells the goods at a retail price of \$570?

The seller earns \$1,849, while the buyer gets \$0, meaning there is no margin for the buyer.

19. What wholesale price would you charge? Please explain why you chose this price.

The main idea is about double marginalization. Setting a wholesale price of \$570 reduces the total supply chain profit by 25% (from \$1,849 to \$1,386.75) because the buyer adds their own markup. The best solution is to set the wholesale price equal to the cost (\$140) and recover profits through franchise fees or revenue sharing. Alternatively, assume if the buyer transparently knows the seller's cost structure, it is fair to split the supply chain profit equally, i.e. set the wholesale price at the midway point between \$140 and \$570, which is \$355.